Digital Logic: Mid Test

Date: 2023. 04. 28 Tue.
Time: 18:30 ~ 21:00

1. (16 pts) Perform the addition operation in 1's complement and 2's complement systems respectively. The number of bits is 8 (8bits binary number), and check if they are overflows or not. (You should convert the numbers to binary ones before addition.)

1) (8pts) 36 - 69

2) (8pts) -33 - 71

2. (14 pts) For an Boolean expression, $AB' + BC + AC'D' + ADF + AEF$,

1) (7pts) Factoring the expression to the minimum product of sum form.

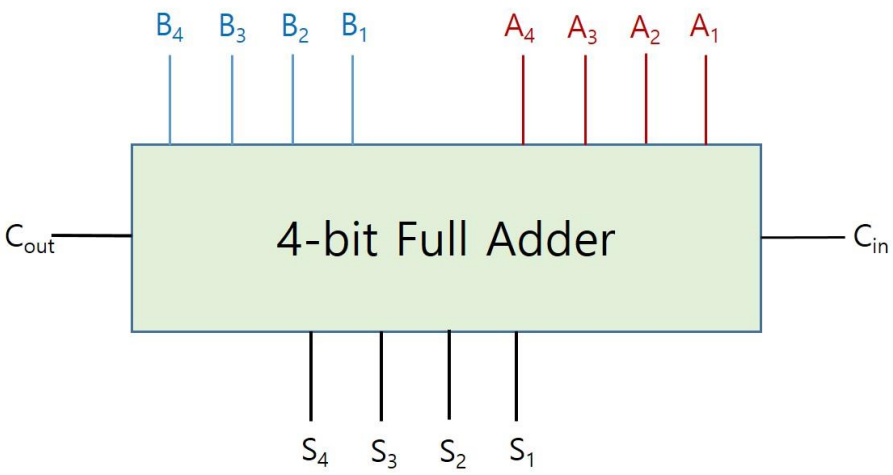
2) (7pts) Multiplying out the expression of result (1) to the minimum sum of product form.

3. (35 points) A function is defined as below,

F(a,b,c,d,e) = ∑m(0,2,8,16,19,20,22,23,26,27,31) + ∑d(10,28,30).

1) (20pts) Find all prime implicants and essential prime implicants respectively using Karnaugh map or Quine-McKluskey method.
2) (10pts) Find all minimum sum-of-product expressions.
3) (5pts) Draw a minimum sum-of-product expression using only NAND gates.

4. (20 pts) Design and draw a multiplier for 4-bit ($B_4B_3B_2B_1$) and 3 bit ($A_3A_2A_1$) unsigned numbers. The output is 6-bit number ($C_6C_5C_4C_3C_2C_1$). You should use **two** 4-bit Full adders, and other AND/OR gates if necessary. Connect the inputs, outputs, and carry in/out exactly.



5. (15 pts) A function is operated as the minimum sum of product form, $A'B' + A'D + BCD + B'CD'$. Implement the function using some OR/AND gates, and one 4-1 MUX. Draw the circuit. You don't have to draw the inverters in the circuit.

6. (30 points) A function is defined as below,

F(A,B,C,D,E) = $\sum$m(0,2,3,5,7,9,11,13,14,16,18,24,26,28,30).

1) (25pts) Find all essential prime implicants using Quine-McKluskey method and prime implicant chart. You should write all the columns of QM method.

2) (5pts) Find a minimum sum-of-product expression.

7. (30 points) Let's design the comparator of two 3-bit numbers, A= $A_3A_1A_0$ and B= $B_3B_1B_0$.

There are 3 outputs in the comparator.

if A==B, then F1=1, F2=F3=0,

else if A>B, then F2=1, F1=F3=0

else if A<B, then F3=1, F1=F2=0

Find the Boolean algebras of F1 and F2, respectively. You don't have to make the minimum sum of product form. Just verify your answer.